※ Topology Recap.

def. (Topology)

Given a set $X(\neq \phi)$, we say $\mathcal{T} \subseteq \mathcal{P}(X)$ is a *topology* on X.
if $\mathcal{T}$ satisfies the following ①, ②, ③ :

① $\phi, X \in \mathcal{T}$

② If $\{E_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}$, then $\bigcup_{\alpha \in I} E_\alpha \in \mathcal{T}$.

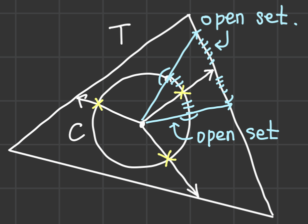
③ If $\{E_k\}_{k \in [N]} \subseteq \mathcal{T}$, then $\bigcap_{k \in [N]} E_k \in \mathcal{T}$.

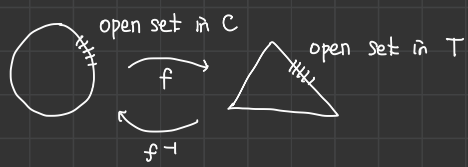
We will call the element of $\mathcal{T}$ the open set.

Rmk. (연속성과 Topology의 관계.)

$f : C \to T$

$f^{-1} : T \to C$





We will call f (which maps the open set in C to the open set in T) the continious function.

def. (Topological space)

If a set X is equipped with a topology $\mathcal{T}$, we will call $(X, \mathcal{T})$ a topological space.

Ex. (i) $(X, \mathcal{P}(X))$ is topological space. (every subset of X is open set.)
: called the discrete topological space. (각 원소 $\{x_1\}, \{x_2\}, \cdots$ 이렇게 존재할 수 있어서 discrete.)

(ii) $(X, \mathcal{T})$, $\mathcal{T} = \{\phi, X\}$ (the most minimal topological space.)
: called the indiscrete topological space. (항상 묶여다녀야해서 indiscrete.)

def. Given a set X, let $\mathcal{T}, \mathcal{T}'$ be topology on X.
We say $\mathcal{T}'$ is finer (larger) than $\mathcal{T}$ if $\mathcal{T} \subseteq \mathcal{T}'$.
$\mathcal{T}'$ is coarser (smaller) than $\mathcal{T}$ if $\mathcal{T} \supseteq \mathcal{T}'$.

In classical analysis, we use the metric topology.

def. (metric)

Given a set X, $d : X \times X \to \mathbb{R}$ is a metric if

① $\forall x, y \in X$, $d(x, y) \geq 0$ and $d(x, y) = 0$ iff $x = y$

② $\forall x, y \in X$, $d(x, y) = d(y, x)$

③ $\forall x, y, z \in X$, $d(x, y) \leq d(x, z) + d(z, y)$

def. (open set on metric space.)

Given (X, d)

① Given $\varepsilon > 0$, $p \in X$, $N_\varepsilon(p) = \{ x \in X : d(x, p) < \varepsilon \}$ called the ε- neighborhood of p.

② We say that $\mathcal{U}$ is open in X if
$\forall p \in \mathcal{U}$, $\exists \varepsilon_p > 0$ s.t. $N_{\varepsilon_p}(p) \subseteq \mathcal{U}$.

Thm. Given (X, d), define $\mathcal{T}_d = \{ \mathcal{U} \subseteq X : \forall p \in \mathcal{U},\ \exists \varepsilon_p > 0 \text{ s.t. } N_{\varepsilon_p}(p) \subseteq \mathcal{U} \}$

Then, $\mathcal{T}_d$ is topology on X.

: called the metric topology. (generated by given metric "d")

: so, it is reasonable to say that $\mathcal{U}$ is open.

Rmk. $\lim_{n \to \infty} a_n = a$

$\iff \forall \varepsilon > 0,\ \exists N_\varepsilon \in \mathbb{N}$ s.t. $n \geq N_\varepsilon \Rightarrow d(a_n, \alpha) < \varepsilon$

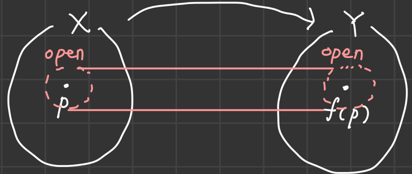
$\iff \forall \varepsilon > 0,\ \exists N_\varepsilon \in \mathbb{N}$ s.t. $n \geq N_\varepsilon \Rightarrow a_n \in N_\varepsilon(\alpha)$

def. (continuity of function.)

A function $f : E (\subseteq \mathbb{R}) \to \mathbb{R}$ is continious at $p \in E$ if,

$\forall \varepsilon > 0,\ \exists \delta > 0$ s.t. $|x - p| < \delta \Rightarrow |f(x) - f(p)| < \varepsilon$

$\iff \forall \varepsilon > 0,\ \exists \delta > 0$ s.t. $x \in N_\delta(p) \Rightarrow f(x) \in N_\varepsilon(p)$



def. (topological view of sequence.)

Given a topological space $(X, \mathcal{T})$,

① A sequence $f : \mathbb{N} \longrightarrow X$ is a function from $\mathbb{N}$ to X.
$$n \longmapsto a_n$$

② Let $\alpha \in X$, $(a_n)_{n=1}^{\infty} \subseteq X$ be sequence in X.
We say $(a_n)_{n=1}^{\infty}$ converges to X if
$\forall$ open set V containing α, $\exists\ N = N(V) \in \mathbb{N}$ s.t. $n \geq N \Rightarrow a_n \in V$.
In this case, denote $\lim\limits_{n\to\infty} a_n = \alpha$.

Rmk. open set V containing α $\equiv$ V is the neighborhood of α.

Rmk. 수열의 수렴성은 위상공간에서 어떤 위상을 선택하는지에 따라 달라진다.
$X = \mathbb{N}$, $\mathcal{T} = \mathcal{T}_c = \{ U \subseteq \mathbb{N} : U^c \text{ is finite} \} \cup \{\phi\}$
Define $a_n := n \quad \forall n \in \mathbb{N}$.

Claim : a_n converges to every points in $\mathbb{N}$.
Take any $m \in \mathbb{N}$.
Take any neighborhood V of m.
Then, V^c must be finite. Thus we can define $N = \max V^c + 1$.
Then, $\forall n \geq N$, $n \notin V^c$. i.e. $n \in V$. $\blacksquare$

open set에 따라 비직관적인 결과가 도래할수 있다.
위 상황을 피하기 위해 "좋은" 위상공간을 고르려고 노력한다. 그럼 좋다는게 뭘지?
$\longrightarrow$ "Hausdorff axiom."

Hausdorff axiom을 만족하는 위상공간 위의 수열은 수렴의 유일성이 보장됨으로 알려져 있다. (위에서는 모든 점으로 수렴.)

* Measure Theory.

def. (sigma-algebra.)

X : set

If $m \subseteq P(X)$ satisfies

① $X \in m$

② If $A \in m$ then $A^c \in m$

③ If $\{A_n\}_{n=1}^{\infty} \subseteq m$ then $\bigcup_{n=1}^{\infty} A_n \in m$

We call an element of σ-algebra a "measurable set."

* "Sigma"라는 term은 countable을 지칭할때 종종 사용된다.

* 통계학에서는 sample space의 sigma-algebra를 고려하고, 그에 대한 measurable set을 "event"라고 지칭한다.

Rmk. (i) ③ 정의에서 countable을 finite로 제한하면 "algebra"가 된다.

(i.e. If $\{A_n\}_{n=1}^{k} \subset m$ then $\bigcup_{n=1}^{k} A_n \in m$.)

(ii) By ①, $X \in m$

By ②, $X^c = \phi \in m$.

Let $A, B \in m \Rightarrow A^c, B^c \in m$.

Then $A^c \cup B^c \cup \phi \cup \phi \cup \cdots \cup \phi \in m$

$= A^c \cup B^c$

Again, $(A^c \cup B^c)^c = A \cap B \in m$. "closed under intersection!!"

Moreover, If $\{A_n\}_{n=1}^{\infty} \subset m$, then $\bigcap_{n=1}^{\infty} A_n \in m$.

If $A, B \in m$, then $A - B \in m$ $(A - B = A \cap B^c)$

(iii) In general, $\exists$ topology $\mathcal{T}$ on some set X s.t. $|\mathcal{T}| = |\mathbb{N}|$

However, $\nexists$ σ-algebra m of X such that m is countable.

(topology와 σ-algebra는 유사해보이나 다르다.)

def. Given m σ-algebra of X and topological space $(Y, \mathcal{T})$,

we say that $f : X \to Y$ is a measurable function

if $\forall\, \mathcal{U} \in \mathcal{T}$, $f^{-1}(\mathcal{U}) \in m$.

Rmk. Let
$$\begin{tikzcd} X \arrow[r, "f", "\text{measurable}"'] \arrow[rr, bend right, "g\circ f"'] & Y \arrow[r, "g", "\text{continious}"'] & Z \end{tikzcd}$$

「continious 와 measurable 함수의 합성은 measurable 이다.」

By definition, $g\circ f$ is measurable. ($\because (g\circ f)^{-1}(\mathcal{U}) = f^{-1}(g^{-1}(\mathcal{U})) \in m$)

Proposition.

Given measurable space (X, m), topological space $(Y, \mathcal{T}_Y)$ and $u, v : X \to \mathbb{R}$ be measurable.

Let $\Phi : \mathbb{R}\times\mathbb{R} \to Y$ be continious.

Define $h : X \to Y$, $x \mapsto \Phi(u(x), v(x))$

Then, h is measurable.

pf. It sufficies to show that $x \overset{f}{\longmapsto} (u(x), v(x))$ measurable.

Let $(a_1, b_1)\times(a_2, b_2) \in \mathbb{R}\times\mathbb{R}$.

Then $f^{-1}(a_1, b_1)\times(a_2, b_2) = \underbrace{u^{-1}(a_1, b_1)}_{\in m} \cap \underbrace{v^{-1}(a_2, b_2)}_{\in m} \in m$.

Rmk. Let $E\in m$. (i.e. E is a measurable set.)

Define $\chi_E(x) = \begin{cases} 1 & \text{if } x\in E \\ 0 & \text{if } x\notin E \end{cases}$

$\chi_E : X \to \mathbb{R}$ is measurable.

정의대로 확인하면 된다. $\forall B\subset \mathbb{R}$, $f^{-1}(B)\in m$ 인지 check.

$$\chi_E^{-1}(B) = \begin{cases} X & 0\in B,\ 1\in B \\ E & 0\notin B,\ 1\in B \\ E^c & 0\in B,\ 1\notin B \\ \emptyset & 0\notin B,\ 1\notin B. \end{cases}$$

한편, by definition, $E, E^c, \emptyset, X \in m$. ▨

prop. Let $\mathcal{F} \subseteq P(X)$.

Then, $\exists$ smallest σ-algebra m^* of X which contains $\mathcal{F}$.

그냥 $\mathcal{F}$를 포함하는 σ-algebra는 trivial 하게 존재한다.

pf. Take $m^* := \bigcap_{\mathcal{F} \subset m} m$

Need to check (i) m^* is σ-algebra

(ii) m^* is smallest.

def. (Borel σ-algebra.)

Given $(X, \mathcal{T})$ topological space.

We call m^* the smallest σ-algebra containing $\mathcal{T}$

the Borel σ-algebra $\mathcal{B}$ of X. ($m^* = \bigcap_{\mathcal{T} \subseteq m} m$)

The element of Borel σ-algebra is called a Borel set.

Rmk. ① Every open set is a Borel set.

② Every closed set is a Borel set.

③ $\bigcap_{n=1}^{\infty} U_n$ & $\bigcap_{n=1}^{\infty} F_n$: Borel set. where U_n open, F_n closed.